

JavaScript & Node.js Lab

Answer by writing the output of each snippet.

Question 1

```
const arr=[1,2,3,4];
const result=arr.filter(x=>x%2===0).map(x=>x*3).reduce((a,c)=>a+c,0);
console.log(result);
```

Question 2

```
function calc(a,b,op){return op(a,b);}
const sum=(x,y)=>x+y;
const multiply=(x,y)=>x*y;
console.log(calc(2,3,sum));
console.log(calc(2,3,multiply));
console.log(calc(5,2,(a,b)=>a-b));
```

Question 3

```
const
user={name:'Ahmed',normal(){console.log(this.name);},arrow:()=>console.log(this.name),nested(){const inner=()=>console.log(this.name);inner();}};
user.normal();
user.arrow();
user.nested();
```

Question 4

```
function counter(){let count=0;return function(){count++;console.log(count);}}
const c1=counter();
const c2=counter();
c1();c1();c2();c1();c2();
```

Question 5

```
const numbers=[1,2,3,4];
const
output=numbers.map(x=>x%2===0?x*2:x).filter(x=>x>2).reduce((a,c)=>{a.push(c*2);return a;},[]);
console.log(output);
console.log(numbers);
```

Question 6

```
console.log("A");
setTimeout(()=>console.log("B"),0);
console.log("C");
```

Question 7

```
console.log("1");
Promise.resolve().then(()=>console.log("2"));
setTimeout(()=>console.log("3"),0);
console.log("4");
```

Question 8

```
console.log("Start");
process.nextTick(()=>console.log("nextTick"));
Promise.resolve().then(()=>console.log("Promise"));
setTimeout(()=>console.log("Timeout"),0);
console.log("End");
```

Question 9

```
console.log("1");
setTimeout(()=>{console.log("2");Promise.resolve().then(()=>console.log("3"));
process.nextTick(()=>console.log("4"));},0);
Promise.resolve().then(()=>{console.log("5");process.nextTick(()=>console.log(
"6"))});});
process.nextTick(()=>console.log("7"));
console.log("8");
```

Question 10

```
const fs=require("fs");
console.log("A");
setTimeout(()=>console.log("B"),0);
setImmediate(()=>console.log("C"));
Promise.resolve().then(()=>console.log("D"));
process.nextTick(()=>console.log("E"));
fs.readFile(__filename,()=>{console.log("F");process.nextTick(()=>console.log(
"G"));Promise.resolve().then(()=>console.log("H"));setTimeout(()=>console.log(
"I"),0);setImmediate(()=>console.log("J"));});});
console.log("K");
```

Bonus 1

```
console.log("1");
setTimeout(()=>{console.log("2");Promise.resolve().then(()=>{console.log("3");
setTimeout(()=>console.log("4"),0);});},0);
Promise.resolve().then(()=>{console.log("5");return
Promise.resolve();}).then(()=>console.log("6"));
process.nextTick(()=>console.log("7"));
console.log("8");
```

Bonus 2

```
const fs=require("fs");
console.log("Start");
fs.readFile(__filename,()=>{console.log("File");setTimeout(()=>console.log("Ti
```

```
meout"),0);setImmediate(()=>console.log("Immediate"));Promise.resolve().then((
) => console.log("Promise1")).then(() => console.log("Promise2"));process.nextTick
(() => console.log("NextTick"));});
Promise.resolve().then(() => console.log("Global Promise"));
process.nextTick(() => console.log("Global NextTick"));
console.log("End");
```